Probability of Default Modeling

Effects of Serial Correlation in Observed Default Rates

Andrija Djurovic

www.linkedin.com/in/andrija-djurovic

Probability of Default Modeling

- Probability of Default (PD) modeling is considered a cornerstone of credit risk modeling.
- Within the credit risk domain, PD models are used for both regulatory and managerial purposes. Among the regulatory applications, the most prominent are IFRS9 and the IRB frameworks.
- Certain PD modeling tasks require the use of time series of observed default rates.
- During PD modeling, practitioners often rely on a range of assumptions. These may include expert-based assumptions regarding specific risk drivers, as well as assumptions related to the statistical methods employed for particular modeling tasks.
- Among statistical assumptions, one of the most commonly used is the assumption of independence of observed default rates over time.
- Given the regulatory requirement that statistical methods used for modeling should be appropriate and sound, as well as the potential consequences of an incorrect independence assumption, it is reasonable to question whether this assumption should be generally accepted in practice or whether practitioners should make additional efforts to properly account for dependence within the relevant modeling procedures.

Assumption of Independence in Observed Default Rates

- Specific areas of PD modeling require analyzing observed default rates over time.
- Typical examples of observed default rate analysis over time within PD modeling include:
  - forward-looking (FLI) modeling under IFRS9;
  - Margin of Conservatism (MoC) Type C estimation under the IRB framework;
  - multi-period backtesting exercises for both IFRS9 and IRB models.
- Therefore, critical aspects of both model development and validation rely on the analysis of observed default rates.
- When performing these modeling tasks, practitioners often rely on the assumption of independence in observed default rates. However, the consequences of an inappropriate assumption of independence may affect different modeling tasks in different ways.
- Moreover, in practice, the assumption of independence is typically considered acceptable, with one of the common arguments being that modeling procedures based on this assumption are more conservative.

Assumption of Independence in Observed Default Rates cont.

- For instance, in Ordinary Least Squares (OLS) regression methods typically used in FLI modeling, the consequences of the dependence structure primarily affect statistical inference, but in certain model designs, they are not a critical aspect of the exercise. In simple terms, the estimated OLS coefficients remain unbiased, while statistical inference related to these coefficients becomes unreliable in the presence of serial correlation in observed default rates. Consequently, practitioners typically design the overall modeling framework by appropriately balancing the criteria used for FLI model selection and validation.
- In the IRB framework, for example, in certain approaches to MoC Type C quantification or multi-period backtesting, serial correlation in observed default rates directly affects the standard error of the estimates, thereby having a significantly greater impact on the overall reliability of the modeling framework.
- These observations challenge the generally accepted assumption of independence and raise the question: if statistical procedures capable of addressing this phenomenon are available within the modeling framework, why is it often not properly accounted for and instead still treated as an acceptable assumption?
- Being aware of the potential impact of an inappropriate independence assumption and being able to quantify its effect on a given modeling task would significantly improve the overall methodological framework for PD modeling, while also motivating further enhancements to statistical methods currently considered best practice.

Simulation Study

Simulation Purpose

- The purpose of this simulation is to demonstrate how an inappropriate treatment of the independence assumption can lead to significantly different outcomes in a given modeling exercise.
- Specifically, the simulation illustrates one approach to quantifying MoC Type C related to the uncertainty of the Long-Run Average (LRA) and examines its relationship with data frequency under both the assumption of independence and an approach that properly incorporates the dependency structure of observed default rates.

Simulation Assumptions

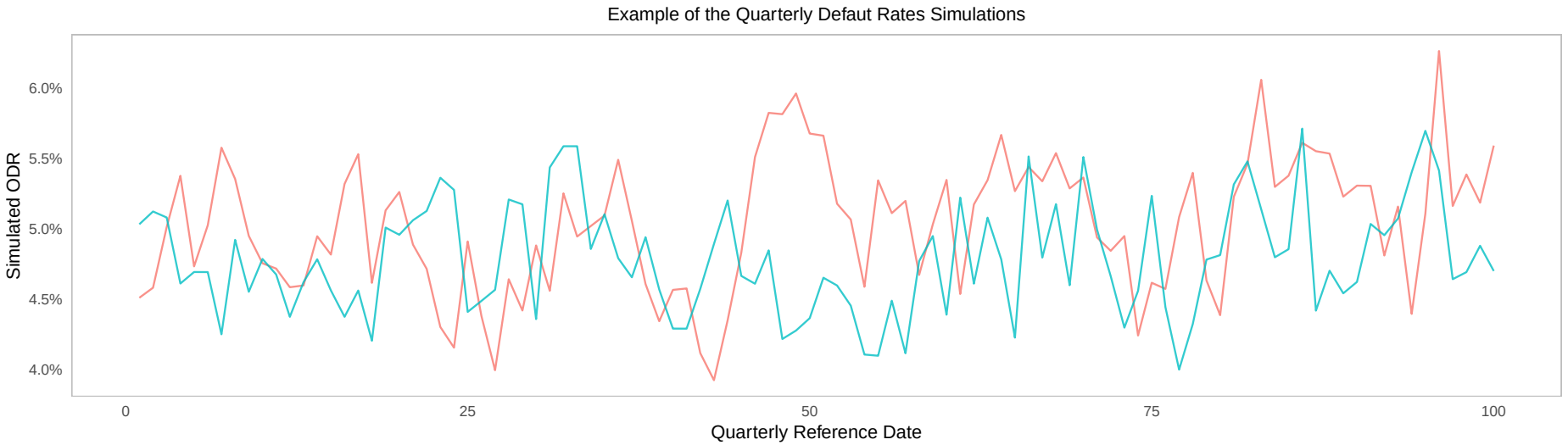
- Assume that, for the purpose of MoC Type C quantification, practitioners are interested in the standard error of the LRA, which can be naively expressed as $\frac{\sigma}{\sqrt{n}}$ where σ denotes the standard deviation of observed default rates and n is the number of data points considered for the LRA calculation.
- The above formula is valid under the assumption that observed default rates are independent across the n time periods.
- Under this assumption, changing the data frequency from annual to quarterly increases the number of observations by a factor of four, leading to a substantially lower estimate of the LRA standard error. However, is this intuitive expectation regarding the effect of a larger sample size reasonable and statistically valid?
- As will be demonstrated by the simulation results, increasing the data frequency while properly accounting for serial dependence in observed default rates does not lead to the same conclusions implied by the naive assumption that a larger number of data points automatically results in a more precise estimate.

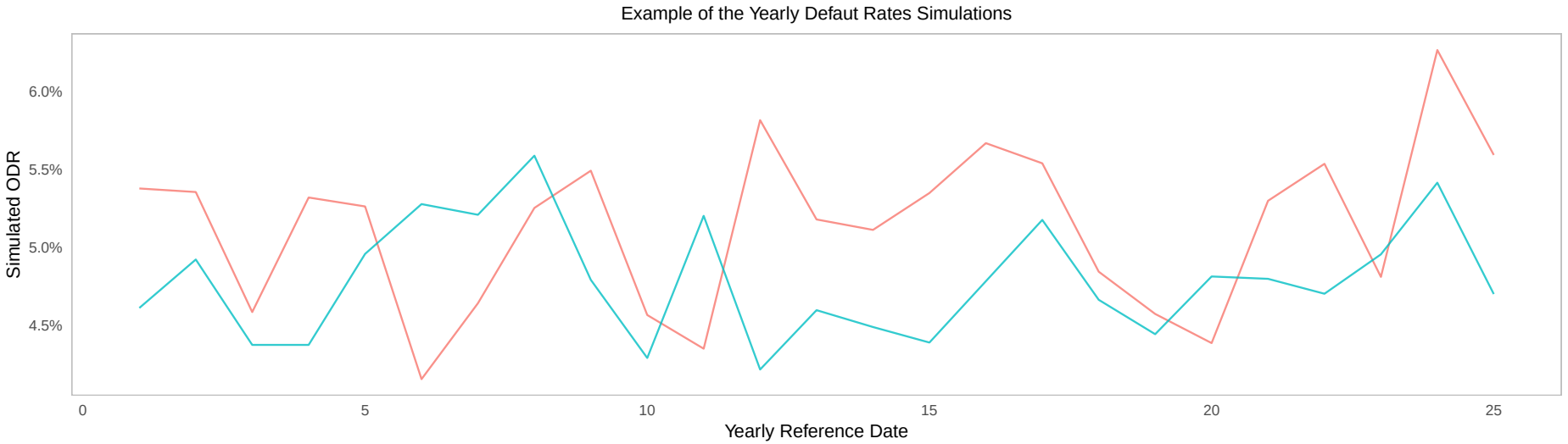
Simulation Study cont.

Simulation Design

- To demonstrate the relationship between sample size and serial correlation, assume that quarterly observed data follow an autocorrelation structure of order four with the following coefficients: 0.55825, 0.13398, 0.01489, 0.00372. Given this quarterly autocorrelation structure, the implied first-order annual (i.e., between year-end observations, or every fourth quarter) autocorrelation coefficient is approximately 0.30.
- Given the assumption of quarterly data frequency and the fourth-order autocorrelation structure, the following steps outline the simulation framework:
  1. for the assumed autocorrelation structure, simulate 100 quarterly default rates from the normal distribution with mean 0.05 and standard deviation of 0.005;
  2. select every fourth observation to obtain 25 observed default rates at annual (year-end) frequency;
  3. calculate the LRA as the average of the quarterly and annual simulated default rates from steps 1 and 2;
  4. repeat steps 1 to 3 a total of 10,000 times and store the resulting averages;
  5. calculate the standard deviation of the quarterly and annual averages;
  6. compare the simulation results from step 5 with the naive standard error of the LRA under the assumption of independence and with the theoretical standard error that accounts for the effective sample size based on the assumed autocorrelation structure.
- The correction factor used for the effective sample size in the calculation of the theoretical standard error is given by $1 + 2\sum_{k=1}^{n-1}\left(1 - \frac{k}{n}\right)\rho_k$ where ρ_k denotes the autocorrelation function at lag k.

Simulation Results





Simulation Results cont.

```
##                    Method se_quarterly se_yearly
##                 Simulated      0.00127   0.00136
##  Theoretical (ACF-based)      0.00127   0.00134
##                     Naive      0.00050   0.00100
```